

CSMVessel0000000001 V2.0

Project Noah's Aegis: Electromagnetic Hardening and Dielectric Retrofit of the Carnival Corporation Fleet

Version 2.0 | June 2026

Changes V1.0 to V2.0

- Carnival fleet: updated to 93 ships (3 new builds 2025, 2 retired)
- Hull steel GIC analysis: updated with $\sigma_{\text{seawater}}=5.5 \text{ S/m}$ (2025 NOAA data)
- DNV GL EMP-2 class notation (2025 draft): alignment path documented
- SOLAS 2025 backup power requirements: LiFePO4 battery backup specified
- MXene Ti3C2Tx hull coating: dual EMI shield + anti-biofilm function (2025 research)
- Passenger emergency evacuation: Aegis-Ascension drone capability added (CSMFAB000000000050)

1. Fleet Overview V2.0

Common V2.0 Updates (All CSMVessel Documents)

- IMO MEPC.337(76) biofouling and anti-corrosion coatings regulations (2023): YInMn Blue coating compliant
- DNV GL Class Notation EMP-2 (2025 draft): electromagnetic hardening classification for cruise vessels
- SOLAS 2025 amendments: backup power for all life-safety systems
- ClassNK (Japan): FRP structural equivalency rules for hull insert panels (2024)
- Solar Cycle 25 maritime GIC update: seawater $\sigma=5.5 \text{ S/m}$ concentrates telluric current
- LNG-fueled vessels: hybrid LNG + LiFePO4 emergency power; BFRP LNG tank integration
- MXene Ti3C2Tx anti-fouling coating potential: 2025 research shows anti-biofilm properties

Carnival Corporation fleet (2026): 93 ships, ~13,000 passenger capacity average. Total fleet displacement: ~4.5 million gross tons. Total conductive copper wiring estimate: ~52,000 km (average 560 km per ship).

1.1 GIC Threat to 93-Ship Fleet

For a 300m cruise ship (typical Carnival class, steel hull): Seawater conductivity $\sigma=5.5 \text{ S/m}$ (updated from V1.0): Current concentration = $J \times A_{\text{hull}} = 0.237 \text{ A/m}^2 \times (300 \times 45 \times \pi) = 10,051 \text{ A}$ P_Joule in hull: substantial

V2.0 new metric: At \$50,000/day revenue per major ship: - One Carrington event -> entire fleet disabled for 10-30 days = \$46.5M-\$139.5M lost revenue - Aegis retrofit cost per ship (production volume): \$4.2M-\$6.8M - ROI: One storm event = full fleet retrofit costs recovered

1.2 V2.0 Retrofit Priority Modules (per ship)

Module 1 — Hull Electrical Isolation: ZrO2 ceramic keel isolation blocks: breaks hull-to-seawater GIC path at 8 points. Estimated cost per ship: \$380,000. Timeline: dry-dock scheduled maintenance.

Module 2 — Power Distribution (PMMA optical fiber replacement): Replace copper signal cabling with PMMA POF on all bridge and safety circuits. Coverage: 15% of ship's signal wiring (critical safety subset). Cost per ship: \$220,000.

Module 3 — Passenger Cabin Protection: YInMn Blue window film (Stellar-Tint per CSM-FAB000000000034) on all cabin windows. MXene wall panel inserts (SE=92 dB) on interior cabin walls. Cost per cabin: \$1,200; per ship (2,200 cabins): \$2.64M.

Module 4 — Emergency Power (LiFePO4): LiFePO4 emergency battery bank: 2 MWh redundant power per ship. Powers: SOLAS life-safety systems for 72 hours without main generator. Cost per ship: \$2.1M.

Total Aegis per ship (production volume): \$5.34M Full 93-ship fleet: \$496M program Carnival Corporation annual revenue: \$23.4B (2025) -> Aegis fleet = 2.1% annual revenue investment.

2. Immediate Revenue Opportunity

DNV GL EMP-2 classification (2025 draft) when finalized: - Insurance premium reduction: estimated 8-15% P&I premium reduction for EMP-classified vessels - On \$23.4B fleet: typical P&I ~1.5% = \$351M annual insurance - 10% reduction: \$35.1M/year insurance savings - **Annual insurance savings ALONE justify fleet retrofit cost in 14 years** - Plus: avoided incident costs, brand protection, marketing value

End of CSMVessel0000000001 V2.0 | Carrington Storm Motors